

NAFIS M. AREFIN

narefin@tamu.edu
linkedin.com/in/narefin
865-770-6705. Houston, TX, USA

EDUCATION	University of Tennessee Knoxville (UTK) <i>PhD, Biomedical Engineering</i> - Advisor: Dr. Bryan C. Good - Dissertation: Embolic Transport Dynamics and Aortic Hemodynamics to Assess Ischemic Stroke Risk	Knoxville, TN, USA 2022 - 2025
	University of Technology Sydney (UTS) <i>MS, Mechanical Engineering</i>	Sydney, Australia 2019 - 2021
	International Islamic University Malaysia (IIUM) <i>BS(Hons.), Aerospace Engineering</i>	Kuala Lumpur, Malaysia 2014 - 2018
RESEARCH INTERESTS	(i) Patient-Specific Cardiovascular Biomechanics - Computational Fluid Dynamics - Fluid-Structure Interaction - Finite Element Analysis - Lagrangian Particle Tracking (ii) Scientific Computing and Predictive Modeling - Deep Learning-Based Image Segmentation - Reduced-Order Modeling - Uncertainty Quantification - Physics-Informed Machine Learning (iii) Image-Based Cardiovascular Digital Twins - FFR-CT Coronary Disease Prediction - Stroke-Risk Prediction - Imaging-Guided Clinical Decision - Personalized Treatment Planning	
EXPERIENCE	Postdoctoral Researcher, Texas A&M University Houston, TX - <i>Cardiovascular Pathomechanics Lab, Texas Medical Center</i> - Advisor: Dr. Lucas H. Timmins (Texas A&M University) - Co-advisor: Dr. Sadeer Al-Kindi, M.D. (Houston Methodist Hospital)	Dec 2025 - Present
	• Developing novel Uncertainty Quantification frameworks in FEA/FSI simulations of coronary artery biomechanics to improve simulation reliability to assess how geometric, material, and boundary-condition variability affects simulation reliability • Implementing deep learning-based CTA image segmentation in Python to characterize segmentation variability & assess its propagation into CFD/FEA digital twins. • Building coupled 3D-0D reduced-order frameworks with Windkessel/ RCR boundary conditions for cardiovascular CFD simulations to capture realistic physiological hemodynamics. • Integrating photon-counting CT images in direct collaboration with clinicians to perform UQ studies aimed at improving patient-specific FFR-CT prediction for coronary artery disease.	
	Graduate Research Assistant, UTK Knoxville, TN • Applied Computational Fluid Dynamics (CFD) , Fluid-Structure Interaction (FSI) , & Lagrangian Particle Tracking (LPT) techniques to simulate emboli transport, aortic hemodynamics, and procedure-related stroke risk during cardiopulmonary bypass. • Processed CT & MRI imaging datasets to segment anatomical structures, reconstruct patient-specific geometries, and generate digital twins for physiologically accurate simulations and predictive modeling. • Designed and prototyped cardiovascular devices and anatomical models using CAD and 3D-printing to support experimental validation and design optimization. • Conducted in-vitro Particle Image Velocimetry (PIV) experiments to capture flow fields, validate computational predictions, and study thrombus formation and growth.	Jan 2022 - Dec 2025

Graduate Teaching Assistant, UTK | Knoxville, TN

Aug 2023 - Dec 2025

- Conducted tutorial and laboratory classes for Fluid Mechanics (AE 341) and Mechanics of Materials sections (ME 321), instructing cohorts of up to 125 UG students a semester.
- Designed and graded examination materials in alignment with course learning objectives and held regular office hours to provide individualized academic support.

Biomedical Design Intern, Medicision | Sydney, Australia

Oct 2020 - Jun 2021

- Designed and tested **3D-printed** miniature CubeSat prototypes for **microgravity cancer cell culture** experiments and spaceflight **biocompatibility**.
- Conducted **CFD & FEA** on CubeSats designed for biomedical applications, focusing on thermal regulation and aerodynamic stability.

R&D Consultant Intern, Optik Consultancy | Sydney, Australia

Jun 2020 - Oct 2020

- Supported the design of **3D-printed dental implants** using CAD modeling and **topology optimization** to improve structural efficiency, manufacturability, and material utilization.
- Conducted **FEA-based biomechanical evaluations** of client-submitted dental implant products to assess deformation, durability, and failure regions under representative dental loading.

Engineering R&D Intern, Bangladesh Airlines | Bangladesh

Jun 2017 - Sep 2017

- Supported Engineering Planning and Services in scheduled aircraft maintenance planning, including C-check and D-checks for **Boeing 737-800 & 777-200 aircrafts**.
- Assisted with **airworthiness-related regulatory compliance** and supported flight trajectory analysis/research to improve aircraft operational efficiency and safety.

PUBLICATIONS

1. **N. Arefin**, M. Usman, A. Narayan, L. Timmins. Uncertainty Quantification in Patient-Specific Vascular Biomechanics Modeling.”. *Journal of Biomechanics* (under review)
2. **N. Arefin**, B. Good. Angled Aortic Cannulation Reduces Cerebral Embolic Transport and Adverse Wall Loading During Cardiopulmonary Bypass: A Patient-Specific Hemodynamic Study. *Cardiovascular Engineering and Technology*
3. **N. Arefin**, B. Good. Emboli Transport in a Full Patient-Specific Aortic Tree: Assessment of Abdominal Organ Injury Risk During Cardiopulmonary Bypass. *Computer Methods and Programs in Biomedicine*, 2026.
4. M. H. Pritom, S. Azad, et al., **N. Arefin**. Efficacy, Safety, and Patient Adherence to AI-Enabled Wearable and Implantable Cardiac Monitoring Devices in Arrhythmia Detection: A Systematic Review. *International Journal of Drug Delivery*, 2026.
5. **N. Arefin**, B. Good. Investigation of cardiopulmonary bypass parameters on embolus transport in a patient-specific aorta. *Biomechanics and Modelling in Mechanobiology*, 2024.
6. M. A. Mohimin, **N. Arefin**, P. Das. Cerebral Aneurysm Hemodynamics in Low Hematocrit Conditions: Numerical Simulations and Analysis. *SSRN, Elsevier*. 2023.
7. S. Khan *et al.*, **N. Arefin**. Assessment of different turbulence models in simulating axisymmetric flow in suddenly expanded nozzles. *International Journal of Engineering & Technology*, 2018.
8. S. Khan *et al.*, **N. Arefin**. Design development of a gravity powered fan using gear transmission. *International Journal of Engineering & Technology*, 2018.

MANUSCRIPTS UNDER PREPARATION

1. **N. Arefin**, B. Good. Cerebral Occlusion Risk during Cardiopulmonary Bypass: A Validated Aorta-to-Circle-of-Willis Hemodynamic Model. *Annals of Biomedical Engineering* (submitted).
2. **N. Arefin**, C. Georgiados, M. Usman, A. Narayan, L. Timmins. Quantifying Deep Learning Segmentation-Driven Geometric Uncertainty in Coronary FFR-CT Predictions.
3. O. Dillenbeck, **N. Arefin**, L. Timmins. Evaluating the Impact of Atherectomy on Arterial Compliance in Patient-Specific Peripheral Artery Disease.

CONFERENCE PRESENTATIONS	1. N. Arefin, B. Good. Cannula-Driven Hemodynamics and Embolic Transport During Cardiopulmonary Bypass Using Validated Patient-Specific CFD-LPT Modeling. <i>10th World Congress of Biomechanics</i>, 2026 (Vancouver, Canada). 2. N. Arefin, B. Good. Cannula-Induced Flow Disturbance and Embolic Risk During Cardiopulmonary Bypass. <i>Cardiovascular Implant Durability Conference</i>, 2025 (California, USA). 3. N. Arefin, B. Good. Computational Investigation of Embolic Injury in Patient-Specific Aortas During Cardiopulmonary Bypass. <i>ASME Summer Bioengineering Conference</i>, 2025 (New Mexico, USA). 4. N. Arefin, B. Good. Computational Analysis of Stroke Risk During Cardiopulmonary Bypass in Adult and Pediatric Anatomies. <i>Summer Biomechanics, Bioengineering, and Biotransport Conference</i>, 2024 (Wisconsin, USA). 5. N. Arefin, B. Good, A. Cripps. The Influence Of Hemodiluted Blood Viscosity On Patient Hemodynamics During Cardiopulmonary Bypass. <i>Summer Biomechanics, Bioengineering, and Biotransport Conference</i>, 2023 (Colorado, USA). 6. M. A. Mohimin, N. Arefin. Cerebral Aneurysm Hemodynamics In Low Hematocrit Conditions: Numerical Simulations and Analysis. <i>14th International Conference on Mechanical Engineering, BUET</i>, 2023 (Dhaka, Bangladesh).		
SKILLS AND TOOLS	CAD: SolidWorks, Blender, SpaceClaim, 3D Slicer, ShapeWorks, CATIA, Fusion 360 CFD/FEA: OpenFOAM, Ansys, LS-DYNA, Abaqus, FEBio, COMSOL, SimVascular Programming: C/C++, MATLAB, Python, PyTorch, TensorFlow, scikit-learn, PyMeshLab		
PROFESSIONAL MEMBERSHIP	American Society of Mechanical Engineers (ASME) American Heart Association (AHA) Biomedical Engineering Society (BMES) Engineers Australia (EA)		
AWARDS AND HONORS	Finalist & Invited Speaker <i>Cardiovascular Implant Durability Conference, CA, USA</i> 2025 Distinguished Research Travel Award <i>Dept. of Biomedical Engineering, UTK</i> 2023, 2025 Summer Research Grant <i>Office of Research, Innovation & Economic Development (ORIED), UTK</i> 2023 Graduate Merit Scholarship <i>Faculty of Engineering and IT (FEIT), UTS</i> 2018 Best Final Year Research Project <i>Faculty of Engineering, IIUM</i> 2018 Dean's List Award <i>Faculty of Engineering, IIUM</i> 2015 Team Runner-Up & Overall Best Speaker <i>National Debating Competition, Malaysia</i> 2015 Gold Medallist & Country Topper (Mathematics) <i>UNSW International Competitions for Schools (ICAS)</i> 2009 1st Runner-up, National Mathematical Olympiad <i>Bangladesh Mathematical Olympiad</i> 2009		

MENTORSHIP AND SUPERVISION	PhD Students	Olivia Dillenbeck Jacob Y. Balster	TAMU BME, 2025–Present TAMU BME, 2025–Present
	MS Students	Christos Georgiadis Maansi Srinivasan	TAMU EnMed, 2025–Present TAMU BME, 2025–Present
	BS Students	Peter Castillo Alison Cripps Amy Chandler	TAMU BME, 2026–Present UTK BME, 2023–2024 UTK BME, 2022